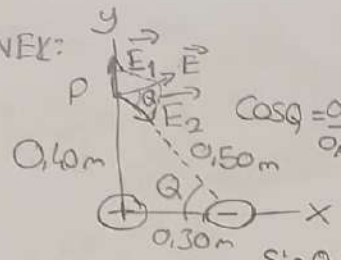
ÖRNEK:




$q_1 = 7{,}0\ \mu C = 7{,}0 \cdot 10^{-6}\ C$

$q_2 = 5{,}0\ \mu C = 5{,}0 \cdot 10^{-6}\ C$

$E = k_e \cdot \frac{|q|}{r^2}$

$k_e = 9 \cdot 10^9\ N \cdot m^2/C^2$

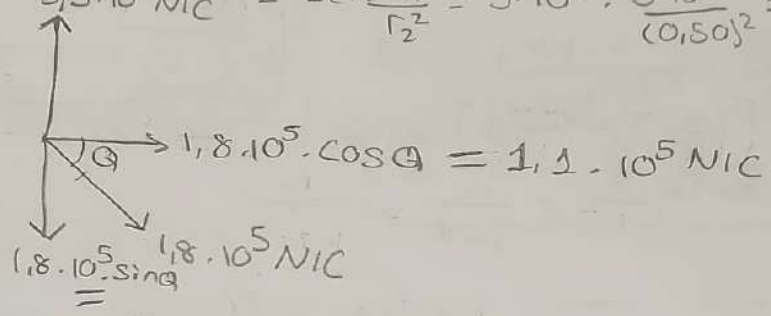
$\cos\theta = \frac{0{,}30}{0{,}50}$

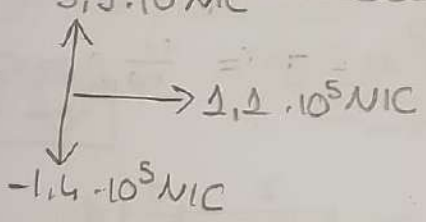
P noktasındaki elektrik alanı bulunuz.

$\sin\theta = \frac{0{,}40}{0{,}50}$

$E_1 = k_e \cdot \frac{|q_1|}{r_1^2} = 9 \cdot 10^9 \cdot \frac{7 \cdot 10^{-6}}{(0{,}40)^2} = 3{,}9 \cdot 10^5\ N/C$

$E_2 = k_e \cdot \frac{|q_2|}{r_2^2} = 9 \cdot 10^9 \cdot \frac{5 \cdot 10^{-6}}{(0{,}50)^2} = 1{,}8 \cdot 10^5\ N/C$




$\vec{E} = 1{,}1 \cdot 10^5\ \vec{i} + (3{,}9 - 1{,}4) \cdot 10^5\ \vec{j}\ (N/C)$

$$\boxed{\vec{E} = 1{,}1 \cdot 10^5\ \vec{i} + 2{,}5 \cdot 10^5\ \vec{j}\ (N/C)}$$

$|\vec{E}| = \sqrt{(1{,}1 \cdot 10^5)^2 + (2{,}5 \cdot 10^5)^2}$

$|\vec{E}| = 2{,}7 \cdot 10^5\ N/C$

$\tan\theta = \frac{2{,}5 \cdot 10^5}{1{,}1 \cdot 10^5}$

$$\boxed{\theta = \tan^{-1}\left[\frac{2{,}5 \cdot 10^5}{1{,}1 \cdot 10^5}\right] = 66^\circ}$$

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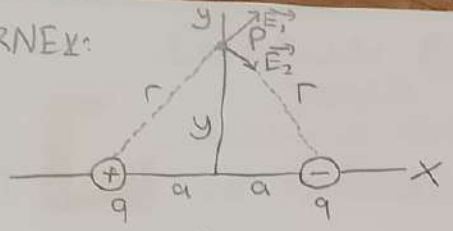
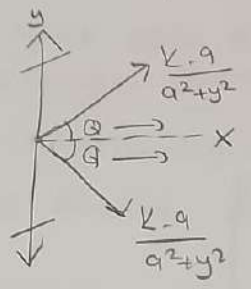
ÖRNEK:




Şekildeki dipol için P noktasından yüklerin oluşturduğu elektrik alan bulunuz. ($y >> a$'dır)




$\vec{E} = \vec{E_1} + \vec{E_2}$ $\quad E_T = 2.E.\cos\theta$

$E_1 = \frac{K.q}{a^2+y^2}$ $\quad E_T = 2 \cdot K_e \cdot \frac{q}{a^2+y^2} \cdot \frac{a}{(a^2+y^2)^{1/2}}$

$E_2 = \frac{K.q}{a^2+y^2}$ $\quad \boxed{E_T = \frac{2 \cdot K_e \cdot q.a}{(a^2+y^2)^{3/2}} \cdot \vec{i}}$

$|\vec{E}| = \boxed{\frac{2 \cdot K_q \cdot q}{(a^2+y^2) \cdot \sqrt{a^2+y^2}}}$

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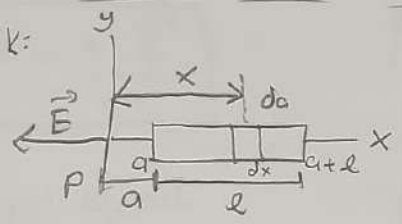
ÖRNEK:




l uzunluklu bir çubuğun toplam yükü Q ve boyca yük yoğunluğu λ'dır. P noktasındaki elektrik alanı hesaplayınız.

$E = K_e \int \frac{dq}{r^2} \Rightarrow E = K_e \int_0^{a+l} \frac{\lambda dx}{x^2}$

$\lambda = \frac{Q}{l} = \frac{dq}{dx}$ $\quad E = \lambda \cdot K_e \left( \int_a^{a+l} x^{-2} \cdot dx \right) \frac{-1}{x}$

$\boxed{dq = \lambda . dx}$

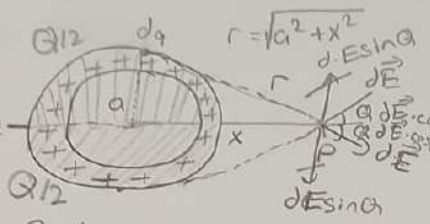
$r \rightarrow x$

$\frac{x^{-2+1}}{-1} = \frac{x^{-1}}{-1} = -\frac{1}{x}$

$t = \lambda \cdot K_e \left( \frac{-1}{x} \right)_a^{a+l} = \lambda \cdot K_e \left[ \left( \frac{-1}{a+l} \right) - \left( \frac{-1}{a} \right) \right]$

$\boxed{E = \frac{K_e (\lambda . l) Q}{a.(a+l)}}$

ÖRNEK: a yarıçaplı bir halka üzerinde düzgün olarak dağılmış Q yükü vardır. P noktasında halkanın elektrik alanını hesaplayınız.




$\cos\alpha = \frac{x}{(a^2+x^2)^{1/2}}$

$E = k_e \int \frac{dq}{r^2}$

$dE = \boxed{k_e \cdot \frac{dq}{r^2}}$

$dE_T = 2 \cdot dE \cdot \cos\alpha$

$dE_T = 2 \cdot k_e \cdot \frac{dq}{r^2} \cdot \frac{x}{(a^2+x^2)^{1/2}}$

$\int dE_T = \int 2 \cdot k_e \cdot \frac{dq}{(a^2+x^2)} \cdot \frac{x}{(a^2+x^2)^{1/2}}$

$E_T = \frac{2 k_e \cdot x}{(a^2+x^2)^{3/2}} \int_0^{Q/2} dq$

$E_T = \frac{2 \cdot k_e \cdot x}{(a^2+x^2)^{3/2}} \cdot \frac{Q}{2}$

$\boxed{E_T = k_e \cdot Q \cdot \frac{x}{(a^2+x^2)^{3/2}}}$

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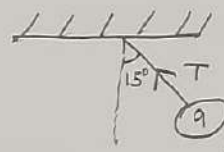
ÖRNEK: Elektriksel kuvvetle itilen yüklü cismin q yükünü hesaplayınız.




$\boxed{m = 2gr}$

$\vec{E} = 1{,}00 \cdot 10^3 \vec{i}$ (N/C)

$T \cdot \sin 15 = 10^3$ $\quad T = 3{,}863 \cdot 10^3$

$T \cdot \cos 15 = m \cdot 9{,}81$ $\quad T \cdot \cos 15 = 3{,}731 \cdot 10^3$

$\frac{T \cdot \sin 15}{T \cdot \cos 15} = \frac{F_e}{m \cdot g}$ $\quad 0{,}267 = \frac{F_e}{2 \cdot 10^{-3} \cdot 9{,}81}$ $\quad \boxed{q = 5{,}25}$

$F_e = 0{,}00525$ $\quad q = \frac{F}{E}$ $\quad q = \frac{0{,}00525}{10^{-3}}$

4. hafta